```java
import java.util.Scanner;

    class Node {
        int data;
        Node prev;
        Node next;

        Node(int data) {
            this.data = data;
            this.prev = null;
            this.next = null;
        }
    }

    class DoublyLinkedList {
        Node head;

        DoublyLinkedList() {
            this.head = null;
        }

        // Function to insert a new node at the end of the doubly linked list
        void insert(int data) {
            Node newNode = new Node(data);
            if (head == null) {
                head = newNode;
            } else {
                Node temp = head;
                while (temp.next != null) {
                    temp = temp.next;
                }
                temp.next = newNode;
                newNode.prev = temp;
            }
```

```java
    }

    // Function to display the doubly linked list
    void display() {
        Node temp = head;
        while (temp != null) {
            System.out.print(temp.data + " ");
            temp = temp.next;
        }
        System.out.println();
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);
        DoublyLinkedList doublyLinkedList = new DoublyLinkedList();

        System.out.print("Enter the number of elements: ");
        int n = scanner.nextInt();

        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            int element = scanner.nextInt();
            doublyLinkedList.insert(element);
        }

        System.out.println("Doubly Linked List elements:");
        doublyLinkedList.display();
    }
}
```